BACKTRACK: competition strategy

The project in one sentence

BACKTRACK gives a learner a new route to a current learning goal: investigate the missing step, practice it with Khan Academy, then demonstrate the return on fresh problems.

The strongest version

The product vision is a recovery navigator for classroom learning and self-study across many curriculum destinations. The current demo supplies five mathematics routes: quadratic equations, equations with brackets, fractions, ratios, and reading linear graphs. The pitch follows one quadratic example so a judge can understand the complete recovery routine without learning the whole product.

The competitive advantage to build is a repeatable recovery workflow, not a supposedly novel adaptive algorithm. A teacher confirms the destination once. A learner sees why a prerequisite is being checked. Evidence removes review or inserts a deeper check. Khan provides the lesson and practice. Fresh destination tasks supply the return evidence. A later check tests whether the recovery holds. Another educator should be able to run this using the same destination kit.

The complete kit is the replication unit: destination definition, reviewed prerequisite graph, discriminating checks, matched Khan resources, parallel return assessments, facilitator guide, and reporting definitions. Its quality and measured usefulness can become an advantage. At present this is a proposed advantage, not an established moat.

The memorable moments

1. **Your GPS recalculates. Why doesn't learning?** Keep the destination fixed while the route changes. The metaphor should take seconds, not a speech.
2. **One review removed.** Two fresh unassisted checks cause an actual stop to leave the visible route. Do not display invented minutes saved.
3. **Back on track, and still there later.** Show a fresh destination problem immediately. In the pilot, earn the later claim with a delayed assessment.

The website supplies an experience that the PDF cannot. Its homepage is short. Its main value is inside the learning flow. The deck carries the case for adoption, implementation, measurement, and continuation.

What changed from the inherited concept

- Removed the claim that general AI tutors cannot diagnose gaps. Gemini study notebooks now do goal-based diagnosis and adaptive learning. ALEKS already includes prerequisites within custom objectives.[1][2]
- Preserved the GPS metaphor as an explanation, not evidence of technical novelty.
- Replaced one-answer prerequisite mastery inferences with two distinct unassisted checks for a route-specific decision. The public product still does not establish durable mastery.
- Added the zero-product step between factoring and solving a quadratic. Being able to factor does not automatically demonstrate knowing what to do with the factors.
- Changed session budgets into manageable blocks. A five-minute choice cannot erase a genuine prerequisite.
- Separated resource opens, self-reported Khan practice, and BACKTRACK answers. Added official Khan videos inside the learning step, with original practice on Khan.
- Chose retained reentry as the signature outcome. Time-to-reentry remains secondary because fast completers alone give a biased efficiency story.
- Replaced the initial rejected visual direction with a predominantly white, Khan-inspired interface, one interface font, restrained accents, and collapsible mobile route context.

How judges should assess the opportunity

The strongest argument is that the team has found an addressable adoption barrier: a learner can have access to excellent material while still not know which earlier step to practice. BACKTRACK creates a specific reason to use Khan now, and a clear reason to return to it. The question is whether that workflow improves retained current-goal performance or reduces wasted effort compared with a simpler Khan routine.

Khan integration carries 25% at application. The proposed teacher assignment and reporting routine matters more than the logo or player. Build the argument around repeated relevant practice, facilitator accountability, and evidence that survives after the session.[3]

Hostile scoring rehearsal

The current review focuses on whether the need, mechanism, and potential are visible in the proposal. The opening explains the learner's difficulty and the complete response. The two-wrong-answers example demonstrates why different help is useful. A focused Khan stop, fresh return task, and later check give the idea a complete learning purpose. These are internal assessments of the entry, not organizer scores or a prediction of rank.

The first review exposed four controllable weaknesses: overly broad novelty claims, weak evidence thresholds, a missing mathematical prerequisite, and confusion between clicks and learning. The revised product addresses those and adds wrong-answer clues, focused in-app repairs, five destinations, and a replayable comparison. The team has now confirmed UP Manila endorsement. Pilot enrollment and learning outcomes remain proposed; institutional endorsement is distinct from completed school implementation.

Why this might not win

An experienced school-adoption team could bring actual teacher commitments, a proven routine, and credible costs. A stronger product team could reproduce a route animation quickly. Khan itself or another adaptive platform could cover the same diagnosis task. A rigorous judge may prefer simple teacher-selected Khan practice if BACKTRACK adds assessment burden without better reentry.

The response is to make the comparison measurable and the school routine light. Before September 18, prioritize teacher review and learner walkthroughs over additional visual features. Before nationals, seek one concrete access agreement and a co-designed routine, with permission before describing any institution as a partner. During implementation, test a second educator's ability to reuse the kit with limited team assistance.

Decision gates

- Before application: verify eligibility, collect actual need evidence where permitted, resolve the executive-summary field discrepancy, and obtain adviser review.
- Before pilot enrollment: approve permissions, destination maps, assessment forms, reporting access, and a feasible school schedule.
- Before expansion: demonstrate usable data, tolerable burden, no unaddressed learner harm, and a replicable workflow. A positive result on a small sample is a signal to test further, not proof of nationwide effectiveness.

Sources

[1] Google, "5 ways to learn with study notebooks in the Gemini app," June 25, 2026. https://blog.google/innovation-and-ai/products/gemini-app/gemini-study-notebooks/

[2] McGraw Hill, "Creating Custom Objectives," undated guide. https://www.mheducation.com/unitas/school/explore/sites/aleks/customizing-custom-objectives.pdf

[3] Enactus Philippines, official KEIC 2026 page, accessed September 10, 2026. https://enactus.ph/2026-national-competition/khan-academy-challenge